function that it models, which will also more effectively guide optimization down the line. We incorporate prior knowledge by selecting appropriate mean and covariance functions for our GP and setting their parameters' values. Specifically,

- The mean function defines the expected behavior of the objective function.
- The covariance function defines the structure of the objective, or more specifically, how quickly and smoothly it changes across its domain.

Each of the choices above leads to drastically different behavior in the resulting GP. For example, a linear mean function will lead to a linear behavior in the GP's predictions, while a quadratic mean function will lead to a quadratic behavior. By using different parameters in the covariance function, we can also control for the variability of our GP.

3.3 Defining the functional behavior with the mean function

First, we will talk about the mean function of a GP, which defines the expected behavior of the GP, or what we believe the function looks like on average, across all possible scenarios of the function. This, as we will see, will help us specify any prior knowledge that is related to the function's general behavior and shape. The code that we will be using throughout this section is included in `CH03/01 - Mean functions.ipynb`.

To make our discussions concrete, we will be using a housing price data set with five data points in Table 3.1.

Table 3.1 Example training data set. The prediction target (price) increases as a function of the feature (living area).

Living area (in squared foot times 1000)	Price (in dollars times 100,000)
0.5	0.0625
1	0.25
1.5	0.375
3	2.25
4	4

In this data set, the function values that we will be modeling is the housing prices, which are strictly positive and increase with larger values of living area. These properties make intuitive sense, and even without knowing what the prices of unobserved houses are, we know for sure that those unseen prices also have these properties.

Our goal here is to incorporate these properties into our mean function, as they describe how we expect the function to behave. Before we jump right to modeling, we first write a helper function that takes in a GP model (along with its likelihood function) and visualizes its predictions in the range from 0 to 10 (that is, 10,000 squared foot in living area). This is implemented as follows:

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